

THE FEDERAL POLYTECHNIC, MUBI
SCHOOL OF HUMANITIES AND CREATIVE ARTS TECHNOLOGY
DEPARTMENT OF LANGUAGES AND COMMUNICATION
2nd AND 4th SEMESTER MAIN EXAMINATION, 2021/2022 ACADEMIC SESSION
5th DECEMBER, 2022

COURSE TITLE: Communication in English III **TIME ALLOWED:** 2hrs 30mins
COURSE CODE: GNS 302

PROGRAMMES: HND I Mass Communication, Civil Engr., Survey & Geo., Crop Production, Agric Extension & Mgt., Public Administration, Library & Inf. Science, Computer Engr., Agric Tech., Mechanical Engr., Computer Science, Chemical Science, Microbiology, Electrical Engr., Biological Science, Animal Health & Production, Agric Bio/Engr. And Environmental Biology,

Instruction: Answer ANY four (4) questions. All questions carry equal marks.

QUESTION ONE:

- a. Define communication.
- b. List any seven (7) principles of letter writing.
- c. Explain any four (4) listed in (1b) above.

QUESTION TWO

- a. Identify any six (6) differences between a Memo and a business letter.
- b. List the nine (9) basic components of a business letter.
- c. Comment on any five (5) components listed in (2b) above.

QUESTION THREE

- a. Define logic.
- b. Briefly explain the following terms in logic:
 - i. Facts
 - ii. Opinion
- c. State whether the following sentences are facts or opinion:
 - i. Nigeria needs intervention.
 - ii. Nigeria has a population of over 200 million,
 - iii. There are many students who are lazy and deceitful.
 - iv. A dog has three legs.
 - v. Camels are found in Sahara regions.

QUESTION FOUR

- a. Define the term project report.
- b. Explain the following features of an introduction (chapter one) in project report.
 - i. Background to the study
 - ii. Statement of the study
 - iii. Scope of the study
 - iv. Significance of the study
 - v. Research questions.

QUESTION FIVE

- a. Define citation in a project report.
- b. Enumerate five (5) functions of a report.
- c. Discuss the following methods of gathering information for a project report:
 - i. Interview
 - ii. Questionnaire
 - iii. Site visit

- iv. Experiment
- v. Observation

QUESTION SIX

Read the following passage and answer the questions that follow:

Former British minister, Mr. John Major said on Thursday, I believe the key to prosperity to our own sub-Saharan Africa depends on the economic success of Nigeria and South Africa. And Nigeria's case the opportunity for growth will be the increasing moves towards stable democracy'.

He noted that Nigeria, being one of the most populous countries in the world must put in place, structures that would guarantee stable policy and reforms that would eradicate poverty.

In a speech entitled "The Youth and Education in a Changing World", he listed global economic recession; political uncertainty in the world's troubled spots as factors that portend great challenges to the youth of this generation.

Major commended the Universal Basic Education (UBE) programme of the Federal government, describing it as the greatest investment that would be repaid in terms of manpower development. He concluded that "Nigeria can be a power house of Africa", as she can be a major avenue for sustainability and prosperity for Africa, not just West Africa but the whole continent. (The Punch Newspaper, Friday February 21, 2003:64).

QUESTIONS:

- a. What is the most important point expressed in this text?
- b. What can you infer from the text is the attitude of Mr. John Major?
- c. Is this speech realistic, sentimental or objective?
- d. What is Nigeria expected to do?
- e. Why is speech narrowed down to Nigeria?

THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

DECEMBER 2022

PROGRAM: *HND1 Electrical Engineering Technology*

COURSE TITLE: *Electrical Power Machine III*

COURSE CODE: *EEP 326*

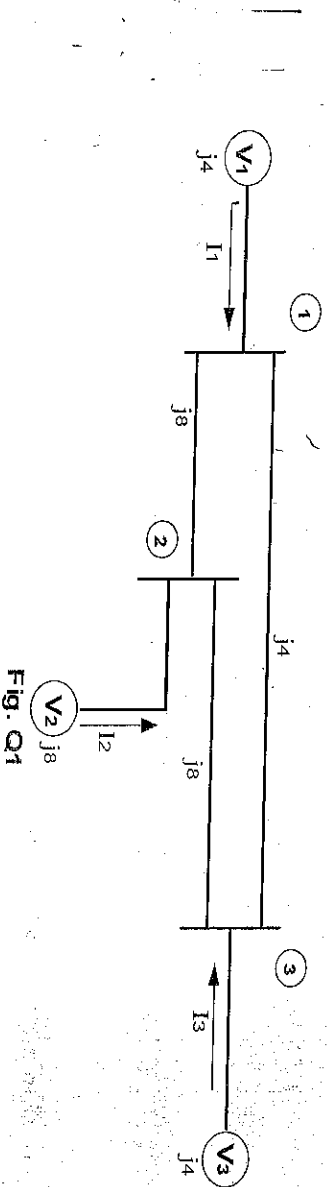
TIME ALLOWED: *3 Hours*

INSTRUCTION: *Attempt any Five (5) Questions only. All question carries equal marks, do not write on the question paper.*

UNIT: 3

QUESTION 1

- Briefly explain bus admittance matrix of power network
- Given the single line diagram of power system in Fig. Q1, obtain the bus admittance matrix by the rule of Inspection.



8mks

12mks

QUESTION 2

- Briefly explain power flow study?
- State the information's that are obtained from a load flow study?
- State the quantities that are associated with each bus in a power network?

5mks

5mks

10mks

QUESTION 3

- List the iterative methods used for solution of load flow problem?
- Given the power network of Fig. Q3, obtain;
 - Y-Bus matrix
 - Bus voltages using Gauss-seidal method with only one (1) iterations
 - Power supplied by Bus 1, S_1

5mks

5mks

5mks

5mks

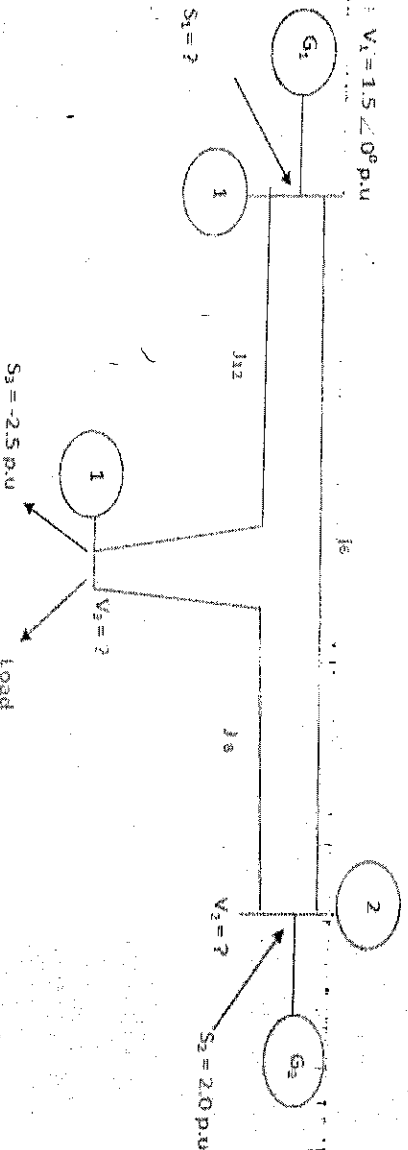


Fig. Q3

QUESTION 4.

- (a) Define transient reactance. **5mks**
- (b) Sketch the waveform of the final stage of fault current of synchronous generator showing the peak values of Sub-transient, transient, and steady state short circuit currents. **10mks**
- (c) A generator rated at 50MVA, 11.8KV has a reactance of 20%. Calculate its P.U. reactance for a base of 100MVA and 10KV? **5mks**

QUESTION 5.

- (a) Briefly explain a fault in power network? **3mks**
- (b) State causes of fault in power network? **5mks**
- (c) Briefly explain fault calculations or fault analysis? **8mks**
- (d) Explain briefly fault level? **4mks**

QUESTION 6.

- (a) Briefly explain symmetrical and unsymmetrical faults? **4mks**
- (b) Define positive, negative, and zero sequence components? **6mks**
- (c) In a 3-phase, 4-wire system, the currents in R, Y, and B lines under unbalanced conditions of loading are as here under:

$$I_R = 120 \angle 30^\circ$$

$$I_Y = 60 \angle 300^\circ$$

$$I_B = 30 \angle 150^\circ$$

Calculate the positive and negative sequence currents.

10mks

QUESTION 7.

- (a) Briefly explain the following:

- (i) Switch gear
- (ii) Circuit breaker

3mks

- (b) State the effects of corona? **3mks**

5mks

- (c) Briefly explain the term voltage surge; sketch a labeled waveform of lightening surge voltage? **3mks**

- (d) Prove mathematically, the surge velocity in a uniform line $3 \times 10^8 \text{m/s}$? **6mks**

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING TECHNOLOGY
END OF 2nd & 4th SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
December, 2022.

PROGRAM: *HND I Electrical and Electronics Engineering Technology.* **UNIT: 2**
COURSE TITLE: *Testing Methods & Reliability.*
COURSE CODE: *EEC 324*
TIME ALLOWED: *2½ Hours*
INSTRUCTION: *Answer two (2) questions from each section. Attempt any Four (4) Questions in all.*
All questions carry equal marks.

QUESTION 1.

(SECTION A)

- (a) Define the following terms? [2 marks each]
 (i) Reliability (ii) Specification (iii) Down time of equipment/system (iv) Up-time of equipment/system
 (b) States any (4) uses of specification [4 marks]
 (ii) Outline the importance of reliability study [3 marks]
 (c) Draw and label the bathtub diagram of equipment/system failure rate analysis? [5 marks]
 (ii) Explain briefly redundancy in reliability study? [1 mark]
 (iii) State any four electronics conditions that may affect equipment/system functionality in reliability study? [4 marks]

QUESTION 2.

- (a) With the aid of a well label diagrams explain briefly the following? [2 marks each]
 (i) Series system arrangement reliability prediction? (iii) Parallel system arrangement reliability prediction?
 (iii) Series-parallel arrangement reliably prediction?
 (b) In order to determine the MTBF of certain components, 100 were tested for period lasting 300 hours. The mean time to failure of the components are shown in the table below. If 70 (seventy) components survive without failure. Assuming the wear out failure is ignored calculate,
 (i) The total test hours before failure [4 marks] (ii) The total test hours without failure? [5 marks]
 (iii) The total survival hours [5 marks] (iv) MTBF [5 marks]

Table 1.

S/N	Number of Components	Time to Failure
1	10	150
2	8	90
3	6	240
4	4	275
5	2	295

QUESTION 3.

- (a) A complex communication satellite system has an in built micro wave repeater unit having a mean time to failure MTTF of 40,000 hrs. The line is operative if only one channel is working and the reliability of the switching unit 0.75. Calculate the reliability of
 (i) a single channel [6 marks]
 (ii) 2 parallel channel [6 marks]
 (iii) 3 parallel channel [6 marks]
 (b) State any six (6) physical conditions that may affects equipment/system functionality in reliability study? [6 marks]

(SECTION B)

QUESTION 4.

(4a) Explain briefly the following maintainability terms?

- (i) MTTR [4 marks each]
- (ii) Maintenance action repair rate (μ) [4 marks each]
- (iii) Availability [5 marks]
- (iv) Unavailability [4 marks]

(b) State five (5) principles objectives of maintenance?

(c) List the two types of planned corrective maintenance?

QUESTION 5.

(a) State five (5) factors that the achievement of high maintainability depends on? [10 marks]

(b) Differentiate between testing and inspection? And states the essence of inspection in industries [10 marks]

(c) List five (5) types of pre-production testing in Engineering firm? [5 marks]

QUESTION 6.

(a) Briefly discuss the following categories of testing and state its importance in Engineering? [6 marks each]

- (i) Production Testing
- (ii) Accelerated Life Testing
- (iii) Reliability Demonstration Testing

(b) State four (4) point to note when or before failure reporting?

(c) Define price definition of maintainability? [4 marks]

[3 marks]

(iv) Unavailability

[5 marks]

[4 marks]

[10 marks]

[10 marks]

[10 marks]

[10 marks]

[6 marks each]

[6 marks]

[6 marks]

[6 marks]

THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

DECEMBER 2022

PROGRAM: *HNDI Electrical Engineering Technology*

COURSE TITLE: *Data Communication & Networking*

COURSE CODE: *ICT 321*

TIME ALLOWED: *3 Hours*

UNIT: 3

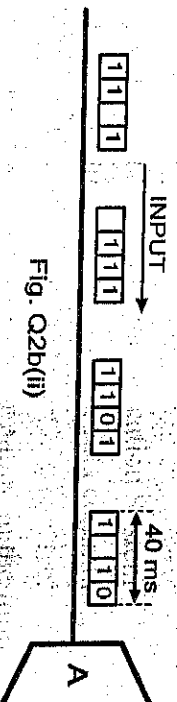
INSTRUCTION: *Attempt any Five (5) Questions only. All question carries equal marks, do not write on the question paper.*

QUESTION 1

- (i) State the function of a Data Communication System. (3 mks)
(ii) Mention the function of the following data communication equipments:
(1) Line Driver (2) Multiplexer. (6 mks)
- (i) Give typical sketch of the flow of information in the following: – (6 mks)
(1) Online Data Collection System (2) Real-time Interactive Processing System. (6 mks)
(ii) List any 2 network monitoring devices and 3 network diagnostic and testing equipments. (5 mks)

QUESTION 2

- (i) Define Amplitude Modulation. (2 mks)
(ii) Give 2 advantages of Phase Modulation over Amplitude and Frequency Modulation. (3 mks)
- (i) State any 2 differences between a Multiplexer and a Concentrator. (4 mks)
(ii) Fig.Q2b(ii), shows input data frames arriving at a multiplexer A. The bits are distributed to the output terminals in the same order of arrival. Use this figure to answer the questions 1 – 10 below. (11 mks)



- What type of multiplexer is A? (2 mks)
- State the mode of operation of A. (3 mks)
- Determine the duration of each input frame in seconds. (3 mks)
- How many input time slots are there? (3 mks)
- Determine the duration of each input time slot. (3 mks)
- Determine the input data rate. (3 mks)
- Determine the number of terminals in the output. (4 mks)
- Determine the output bit duration. (4 mks)
- Determine the output data rate. (4 mks)
- Show the contents of each output terminal. (11 mks)

QUESTION 3

- (i) Define Full Duplex Transmission and give one example. (3 mks)
(ii) Sketch and label the following;
(1) Half duplex transmission. (3 mks)
(2) Serial transmission of the bit stream, 1 1 0 1. (4 mks)
- (i) Mention any 2 differences between Asynchronous and Synchronous transmission modes. (4 mks)
(ii) Show how the 3-character code frames given below can be transmitted by Asynchronous Transmission. (6 mks)

1 1 1 1

0 1 1 0

1 0 0 1

⇒

QUESTION 4

- (a). (i) Define Point-to-point line connection. (4 mks)
(ii) Give typical of sketch Multidrop or Multipoint line or link. (6 mks)
- (b). (i) List all the types of Communication Software we have. (4 mks)
(ii) For each function or location of the communication software given in the table below, state the type of communication software associated with it. (6 mks)

S/N	Function of Communication Software	Type
(1)	Formerly it is located in host computer but now located in the FEP	
(2)	Internet browsers, terminal emulators, WhatsApp	
(3)	Located only in the host computer or server	
(4)	Gathers information about how the network is working	
(5)	Error handling within the host computer	
(6)	Located in switches, routers, and intelligent terminal controllers	

QUESTION 5

- (a). (i) Give 4 reasons why computer networks are needed today. (6 mks)
(ii) Use a table to list all the layers of components in a computer network and give one example for each layer. (6 mks)
- (b). (i) Give the classification of computer networks based on Technological Arrangement. (2 mks)
(ii) Mention any 3 characteristics of a local area network (LAN). (6 mks)

QUESTION 6

- (a). (i) Mention 4 common problems that corrupts data transmission in coaxial and twisted pair cables. (4 mks)
(ii) Give a well labelled typical sketch of S/STP cable. (4 mks)
- (b). (i) Specify any 3 differences between Satellite Microwave and Fibre Optic Cable. (6 mks)
(ii) For each characteristic, advantage and connector described in the table below, mention the transmission medium associated with it. (6 mks)

S/N	Description	Type of Medium
(1)	Uses MT-RJ connectors	
(2)	Ideal for locations where cable installation is difficult or expensive	
(3)	Transmission distance before a repeater is required is 186 metres	
(4)	Twisted wire, shield and screen are applied to protect the data in the cable from deterioration.	
(5)	Does not require any government regulation, licence or permit	
(6)	Easily affected by smoke, fog, and heat wave	

QUESTION 7

- (a). (i) State any 2 reasons why communication protocols are needed. (4 mks)
(ii) State ISO Reference Model for data communication and give a well labelled typical sketch. (4 mks)
- (b). (i) Mention any 2 differences between byte-oriented and bit-oriented protocols. (4 mks)
(ii) Sketch and label a typical frame format for High-Level Data Link Control (HDLC) protocol. (8 mks)

THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING TECHNOLOGY

END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

DECEMBER 2022

PROGRAM: *HND I Electrical Engineering Technology*

COURSE TITLE: *Electrical Machine III*

COURSE CODE: *EEP 327*

TIME ALLOWED: *3 Hours*

UNIT: 3

INSTRUCTION: *Attempt any Five (5) Questions only. All question carries equal marks. Do not write on the*

question paper

1. a.) Define Electromagnetic Induction? (4marks)
b.) Briefly explain the following with respect to Electrical Machines
i. Interaction (4marks)
ii. Alignment (4marks)
c.) The field winding of a D.C electromagnet is wound with 800 turns and has resistance of 40Ω when exciting voltage is 230V. If the magnetic flux density and the cross-sectional area of the coil is 1.2T and 2cm^2 respectively. Calculate
i) the Self-inductance of the coil (5marks)
ii) The energy stored in the magnetic field (3marks)
2. a.i) Describe briefly electromechanical Energy conversion devices. (3marks)
ii) Give the general energy balance equation applicable to all situations. (2marks)
b.i) Draw a schematic diagram of electromechanical energy conversion (2marks)
ii) Deduce the total electrical energy input of electromechanical energy converter. (3marks)
c.) the relay shown in figure 1 is made from infinitely- permeable magnetic material with a movable plunger, also of infinitely permeable material. The height of the plunger is much greater than the air gap length ($h \gg g$). Calculate the magnetic stored energy as a function of plunger position ($0 < x < d$) for $N=1000$ turns, $g=2.0\text{mm}$, $d=0.15\text{m}$, $l=0.1\text{m}$ and $i=10\text{A}$. (Hint: $\mu_0 = 4\pi \times 10^{-7}$). (10 marks)
3. a.) Describe briefly with aid of diagram, Singly Excited Magnetic field system. (4marks)
b.) Draw a model and mention the major parts of singly excited electromechanical energy conversion system. (5marks)
c.) Derive the dynamic equation of singly excited electromechanical system. (11marks)
4. a.) Briefly describe the efficiency of a transformer. (4marks)
b.) How would you determine the efficiency of a transformer? (2marks)
c.) The following reading were obtained from Open circuit and Short circuit test on 8KVA, 400/120V Transformer:
Open Test (L.V side) : 120V, 4A, 75W
Short Circuit Test (H.V side) : 9.5V, 20A, 110W
i.) Determine the Parameters. (9marks)
ii.) Draw the Equivalent circuit. (5marks)
5. a.) i) Describe briefly with the aid of diagram, transformer on Load. (5marks)
ii) Draw and explain briefly the various load phasor diagrams of transformer on Load. (6marks)

- b.) The high voltage and low voltage winding of a 2200/220 transformer has resistance of 4.8Ω and 0.04Ω and reactance 2Ω and 0.018Ω respectively. The low voltage winding is connected to a load having an impedance of $(6+j4)\Omega$.

Determine (i). Equivalent impedance referred to L.V side (3marks)

(ii). Current in L.V winding (2marks)

(iii). Load voltage (2marks)

(iv). Power Consumed by the load. (2marks)

6. a.) With the aid of diagram only show two ways by which single phase transformer can be connected to form a single three phase transformer. (4 marks)
- b.) State the conditions for parallel operation of transformer? (5marks)
- c.) A 1000KVA transformer with a percentage impedance of $(1+j4)$ was installed in Mubi 132/33KV substation to power Mubi metropolis. After the insurgency, there is an increase in load consumption to the tune of 750KVA at 0.8 p.f lagging due to the increase in industrial activities in Mubi. Therefore, an additional 500KVA with a percentage impedance of $(2 + j6)$ was installed. If the no load secondary voltage of each of these transformers is 550V and were connected in parallel, find the load share by each of the transformer and their respective power factor. (11marks)
7. a.) State any six advantages of 3- phase transformer over single phase transformer. (6marks)
- b.) Mention ways of connecting three single phase transformers to form three phase circuits. (4marks)
- c.) the primary winding of a delta-star connected 50VA transformer is supplied with a 100V 50Hz three phase supply. If the transformer has 500 turns on the primary and 100 turns on the secondary winding: calculate the secondary side voltages and current. (10marks)

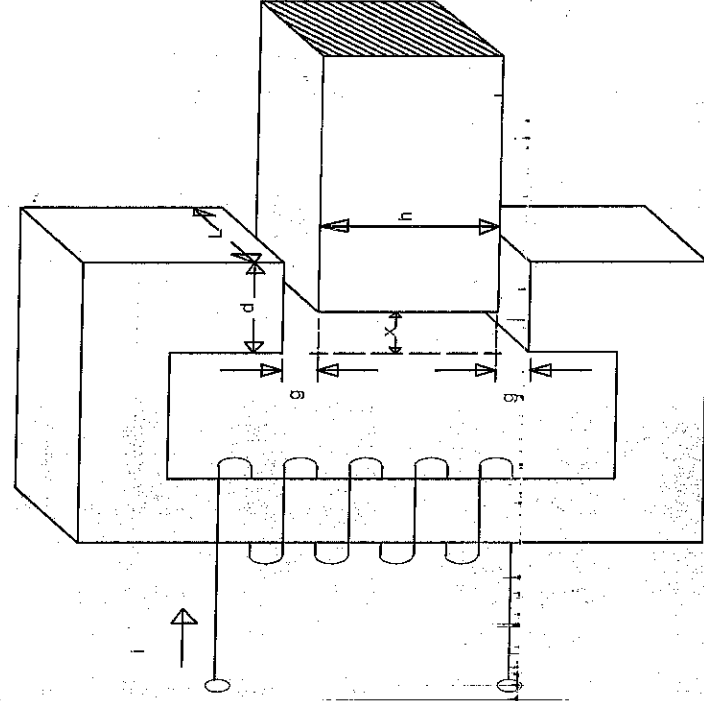


FIGURE 1₂

**THE FEDERAL POLYTECHNIC, MUBI
BUSINESS ADMINISTRATION DEPARTMENT**

2ND & 4TH SEMESTER EXAMINATION 2021/2022 ACADEMIC SESSION

PROGRAM: HND I ELECTRICAL ENGINEERING

COURSE TITLE: INDUSTRIAL MANAGEMENT

COURSE CODE: GNS 413

INSTRUCTION: ANSWER ANY FIVE QUESTION

1. Maintaining profitability means making sure that revenue stays ahead of the cost of doing business, that means every organization follow some steps to achieve organizational objectives; discuss.
2. Management has some functions to perform that span across all industries/businesses as a process where each step builds on the others; justify.
3. The concept of theory X and theory Y was developed by social psychologist Douglas McGregor; elaborate.
4. Write short notes on the following:
 - i. Recruitment
 - ii. Trade union
 - iii. Industrial court
 - iv. Depreciation
 - v. Quality control
 - vi. Market segmentation
 - vii. Product differentiation
5. Money is often defined in terms of the three functions or services that it provides; expatiate
6. Explain the types of financial institutions you know
7. What are the importance of investment management?

GOODLUCK

THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY

END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

DECEMBER 2022

PROGRAM: *HND1 Electrical Engineering Technology*

COURSE TITLE: *Digital Communication I*

COURSE CODE: *EEE 325*

TIME ALLOWED: *3 Hours*

INSTRUCTION: *Attempt any Five (5) Questions only. All question carries equal marks, do not write on the question paper.*

UNIT: 3

QUESTION 1.

- (a) Briefly explain the following (Use diagrams where necessary).
 - (i) Basic Telecommunication Engineering concepts. (4 marks)
 - (ii) Basic Elements of a telecommunication system. (4 marks)
 - (iii) Modes of communication. (4 marks)
 - (iv) Communication System basics. (4 marks)
- (b) Briefly explain continuous time signal and discrete time signal. (4 marks)

QUESTION 2.

- (a) Briefly explain the properties of a signal. (5 marks)
- (b) Draw and show the parameters of a signal. (5 marks)
- (c) (i) Briefly explain Modulation. (5 marks)
(ii) Briefly explain with the aid of a diagram the double sideband modulation. (5 marks)

QUESTION 3.

- (a) Briefly explain Amplitude modulation (use diagrams). (8 marks)
- (b) Given that $V_c(t) = A \cos(2\pi f_c t + \phi)$ derive an equation for the amplitude spectrum of a fully modulated carrier wave which is devoid of distortion or carrier phase reversal. (10 marks)
- (c) Briefly explain modulation index. (2 marks)

QUESTION 4.

- (a) A tuned-oscillator circuit in a simple transmitter employs a 50-micro hertz ($50\mu\text{Hz}$) coil and a Nano farad (1nf) capacitor. If the oscillator is modulated by audio frequency up to 10 KHz, what will be the frequency range occupied by the sidebands? (8 marks)
- (b) If an FM Modulator with a frequency deviation constant gain of 10kHz/V has a sinusoidal modulating signal $M(t) = 4 \cos 2\pi \times 10^3(t)$, calculate
 - (i) The peak frequency deviation. (2 marks)
 - (ii) The modulation index. (2 marks)
 - (c) Briefly explain Demodulation. (8 marks)

QUESTION 5.

- (a) From a diode detector circuit of an AM Demodulator having the following resistances: $R_1 = 100\text{K}\Omega$, $R_2 = 200\text{K}\Omega$, $R_3 = 400\text{K}\Omega$, $R_4 = 900\text{K}\Omega$, calculate the maximum index to be applied to the diode circuit without causing any negative peak clipping. (5 marks)
- (b) Distinguish between frequency modulation and amplitude modulation. (5 marks)
- (c) Briefly explain Fourier series. (5 marks)
- (d) Why do we study Fourier series in digital communication? (5 marks)

QUESTION 6.

- (a) Distinguish between Coherent detector (DSB-AM) and Non-coherent detector in an AM Demodulation. (8 marks)
- (b) Show the mathematical expression of a coherent detector. (5 marks)
- (c) With suitable diagrams briefly explain pulse code modulation. (7 marks)

QUESTION 7.

- (a) Briefly explain Pulse amplitude modulation. (5 marks)

(b) Why do we need modulation in digital communication? (5 marks)

- (c) Briefly explain deviation ratio? (5 marks)

(d) Briefly explain the following.

- (i) A modulator circuits. (1 marks)
- (ii) A detector or mixer circuit. (1 marks)
- (e) Briefly explain Narrow band F.M. (3 marks)

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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION

DECEMBER 2022

PROGRAM: *HND1 Electrical Engineering Technology*

COURSE TITLE: *Electrical Circuit Theory IV*

COURSE CODE: *EEEC 325*

TIME ALLOWED: *2 1/2 Hours*

INSTRUCTION: *Attempt any Five (5) Questions only. All question carries equal marks.*

UNIT: 3

QUESTION 1.

(a) Given the function defined by; $F(t) = 3, 0 < t \leq 2$

$$F_1(t) = 0, 2 < t < 4$$

$$F_2(t) = f(t + 4)$$

(i) Sketch the waveform.

(ii) Find the Laplace transform of the functions.

(b) List two (2) types of a transmission line

(c) Define the following terms,

(i) Wavelength

(ii) Velocity of propagation.

(iii) Propagation constant.

QUESTION 2.

(a) (i) Determine the S-poles and zeros of $H(s) = \frac{s-5}{(s+2)(s-1)}$

(ii) Plot the graph of S-poles and zeros of the function in Q2 (a) above.

(iii) List the procedural steps involved in finding out the poles and zeros of a function.

(b) (i) use the initial and final value theorem to determine the initial and final position of the control value of

$$X_0 \text{ and } X_\infty \text{ where } X_s = \frac{80.8}{s^2 + 2s + 101}$$

Assuming $i(t) = 10mA$, for $t < 0$ and $I(t) = 10mA$, for $t > 0$.

(ii) State the initial and final value theorem.

QUESTION 3.

(a) (i) State first shift theorem.

(ii) Use first shift theorem to find the Laplace transform of $2e^{3t} \sin 3t$.

(b) Find the inverse Laplace transform of the following expressions,

(i) $\frac{1}{s(S^2 - \theta)}$ by partial fraction.

(ii) $\frac{s+2}{s^2 - 5s + 6}$ by Heaviside formula.

(iii) Find the Laplace transform of $F(t) = \sin 2t$.

QUESTION 4.

(a) A transmission line having negligible losses has primary line constants with inductance $L = 0.5mH / \text{loop km}$ and capacitance $C = 0.12\mu F / \text{km}$. Determine the following at an operating frequency of 400kHz.

(i) The characteristics coefficient.

(ii) The propagation coefficient.

(iii) The wavelength of the line.

(iv) The velocity of propagation in meters per second of a signal.

(b) Define the following transmission lines termination,

- (i) Open circuited termination.
- (ii) Short, circuited termination.

(4marks)
(4marks)

QUESTION 5.

- (a) (i) Define Laplace transform.
- (ii) use integration by part to solve the Laplace transform of a function $F(t) = \cos at$
- (b) Find the Laplace transform of the following

(2marks)
(7marks)

- (i) $F(t) = 2 \sin 3t + \cos 3t$
- (ii) $F(t) = 4e^{2t} + 3 \cosh 4t$
- (iii) $F(t) = 2 \sin 3t + 4 \sinh 3t$

(3marks)
(4marks)
(4marks)

QUESTION 6.

- (a) With the aid of sketches, briefly explain the following,
 - (i) Heaviside unit step function.
 - (ii) Dirac-Delta impulse function.
- (b) (i) Use Laplace transform to find the current (I) in the circuit below the switch (S) is closed.

(3marks)
(3marks)

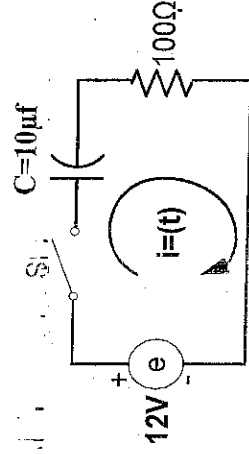


Fig. Q6 (b)

- (ii) State the second shift theorem.
- (iii) Find the inverse of $\frac{Se^{-s}}{s^2 + 9}$ by second shift theorem.
- (iv) Find the inverse whose transform is $\frac{e^{-4s}}{s^2}$ by second shift theorem.

(7marks)
(2marks)
(2½marks)
(2½marks)

QUESTION 7.

- (a) Briefly explain the four (4) transmission primary line constants.

(8marks)

- (b) At frequency of 1kHz, the primary transmission line constant is resistance $R = 25 \Omega$ / loop km, inductance $L = 5 \text{mH}$ / loop km, capacitance $C = 0.05 \mu\text{F/km}$ and conductance $G = 80 \text{mhos} / \text{km}$. calculate for the lines the following

- (i) The characteristics impedance.
- (ii) The propagation coefficient.
- (iii) The attenuation coefficient.
- (iv) phase-shift coefficient.

(3marks)
(3marks)
(3marks)
(3marks)

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 2ND & 4TH SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
DECEMBER 2022

PROGRAM: *HND1 Electrical Engineering Technology*

UNIT: 2

COURSE TITLE: *Electrical Design & Drafting I*

COURSE CODE: *EEP 328*

TIME ALLOWED: *2½ Hours*

INSTRUCTION: *Attempt any Four (4) Questions only. All question carries equal marks.*

1. (a) Define the following terms in relation to electrical installations of a building;
(i) Switchgear (3mks) (ii) Spare ways (3mks) (iii) Fuse (3mks) (iv) Consumer unit (3mks)
(b) Define the following “factors” in relation to electrical design;
(i) Load (2mks) (ii) Utilization (2mks) (iii) Diversity (2mks) (iv) Circuit breaker. (2mks)
(c) A 20 hp machine drives a constant 16 hp load when in operation. If the motor is operated for 8hrs/day for six days in a week. Determine:
(i) Use factor per week (3mks) (ii) Load factor (2mks)
2. (a) Define the term “**Load balancing**” in a secondary distribution network of power supply. *(7mks)*
(b) With the aid of diagram only, connect the following loads on a secondary distribution line of 3-phase 4-wire system, taking load balancing in to consideration.
(i) Load A; Single phase machine connected to blue phase rated 3 hp. (2mks)
(ii) Load B; 3-phase induction motor rated 10 hp. (2mks)
(iii) Load C; A self-contained Apartment rated 4.2 kW. (2mks)
(iv) Load D; A shopping mall rated 20 kW. (2mks)
(c) What are the two shapes accepted for busbar construction and why are they recommended? *(10mks)*
3. (a) Define the following:
(i) Wiring diagram (3mks) (ii) Wiring system (3mks)
(b) List the three basic types of wiring system and briefly explain all. *(9mks)*
(c) Briefly explain the nature of the following wiring diagrams:
(i) Layout (2mks) (ii) Block (2mks) (iii) Logic (2mks) (iv) Schematic (2mks) (v) Highway (2mks)
4. (a) Cables are basically constructed in to three (3) parts, list the parts and state their functions. *(2mks)*
(b) Using Tables 1(a) 7 Table 1(b), determine the minimum PVC conduit size that can contain two *1.5mm², two 4mm², and one 6mm², if the wires to be drawn contains only two bends and runs for 5m long. (2mks)*
(c) A 100m² working plane requires an illumination of 400lx, if an LED lamp rated 100w an possesses an efficacy of 150lm/w is to be used, calculate the minimum number of lamps to be employed, assuming maintenance factor and coefficient of utilization of 0.8 and 0.9 respectively. *(2mks)*
5. (a) Define the following:
(i) Luminous Intensity (2mks) (ii) Illuminance (2mks) (iii) Luminous flux (2mks)
(iv) Luminous efficacy (2mks)
(b) Briefly explain the under listed fitting systems of lighting distribution schemes and also state the percentage of downward and upward distribution in each case: *(i) Direct fitting (3mks)*
(ii) Indirect fitting (3mks) (iii) Semi-direct fitting (3mks)
(c) Draw separate diagrams for each type of fitting listed in Question 5 (b) above showing the physical nature of the lamp installations. *(8mks)*

6. (a) Solar street lamps A, B and C emits 500Cd each in all directions, if the lamps are 4m above the ground plane and all the lamps illuminate a point "y" directly beneath "B" on the ground plane, with lamp "A" placed on the left of lamp "B" and having perpendicular distance of 2m and Lamp "C" is on the right side of Lamp "B" and at a perpendicular distance of 2m. then:

(i) Draw the diagram illustrating lamps positions, ground plane and various distances involved.

(ii) Calculate the illuminance produced at point “y” due to the 3 lamps. (5mks) (10mks)

(b) List the two (2) methods used for estimating illumination levels.

(c) State the following laws using equations only:

(i) Inverse square Law.

(ii) Inverse square Law. (3mks)

(ii) Cosine Law. $(3mks)$

Table 1(a) Conduit cable factors. Adapted from the IEEE

Type of conductor	Conductor cross-sectional area (mm ²)	Cable factor
Solid or stranded	1	18
	1.5	22
	2.5	30
	4	48
	6	58
	10	105
	16	145
	25	217

The lower radius of a conduit bend should be not less than 2.5 times the outside diameter of the conduit.

Table 1(b) Conduit cable factors for bends and long straight runs

Length of run (m)	Conduit diameter (mm)										
	16	20	25	32	40	50	63	75	90	110	125
1	Straight										
1.5	Covered by										
2	Tubes										
2.5	A and B										
3	A and B										
3.5	A and B										
4	A and B										
4.5	A and B										
5	A and B										
5.5	A and B										
6	A and B										
7	A and B										
8	A and B										
9	A and B										
10	A and B										

THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF MATHEMATICS AND STATISTICS 2ND AND 4TH SEMESTER 2021/2022 EXAMINATIONS, DECEMBER 2022.

PROGRAMME: HND I Electrical/Mechanical/Survey/Physics electronic/Agric. Engr./Comp. Engr.

COURSE TITLE: Advanced Calculus COURSE CODE: MTH 312

INSTRUCTION: Answer any four Questions

TIME ALLOWED: 2 $\frac{1}{2}$ Hours

- 1 (a) Define the Laplace transform of a function $f(t)$ and hence or otherwise find the Laplace transform of $f(t) = e^{at}$
(b) A first order differential equation involving current i in a series R-L circuit is given by $\frac{di}{dt} + Si = \frac{E}{2}$ and $i = 0$ at time $t = 0$. Use the Laplace transform to solve for i when $E = 40e^{-3t}$
- 2 (a) When do a function $f(x)$ is said to be even or odd and hence show that $f(x) = x^5 - x^2 + 3$ is even.
(b) Find the half-range Fourier sine series to represent the function $f(x) = 3x$ in the range $0 \leq x \leq \pi$ of period 2π .
- 3 (a) Find the general solution of the homogeneous differential equations
(i) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$ (ii) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$
(b) Solve the non-homogeneous differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2$
- 4 (a) Find the particular solution of the equation $y'' - 2y' - 3y = 0$ when $y(0) = 2$ and $y'(0) = 14$
(b) Solve the simultaneous differential equations $\frac{dx}{dt} + x + 3y = 0$, $2x + \frac{dy}{dt} + 2y = 0$
- 5 (a) (i) Show that the equation $2x(x^3 + y^3)dx + 3y^2(x^2 + y^2)dy = 0$ is exact and hence solve the equation (ii) Solve the equation $e^x \frac{dy}{dx} = 6$, given that $y = 3$ when $x = 0$
(b) Solve the partial differential equation $\frac{\partial^2 u}{\partial x^2} = 12x^2(t + 1)$ given that $x = 0$, $u = \cos 2t$ and $\frac{\partial u}{\partial x} = \sin t$
- 6 (a) (i) Find all second order partial derivatives of $f(x, y) = x^3y - y^3 + \ln x$
(ii) Find all the critical points of $f(x, y) = 2x^2 - y^3 - 2xy$
(b) (i) Solve the Cauchy equation $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - 9y = 0$
(ii) Evaluate the line integral $I = \int_C \{x^2 + 2y\}dx + xydy$ from O (0, 0) to B (1, 4) along the curve $y = 4x^2$

